

Mahimna Darji

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SUMMARY

Data professional with 2 years of experience leading a three-person team and building machine learning, NLP and GenAI solutions. Experienced in data preprocessing, feature engineering, EDA, model training and evaluation using Python, Scikit-learn, TensorFlow and PyTorch across structured and unstructured datasets. Skilled in developing RAG systems, semantic analysis workflows, supervised and unsupervised models, and deploying AI applications using Flask, APIs, Docker and Kafka-based pipelines with basic monitoring and responsible AI awareness. Worked closely with executive leadership, including the Lead Data Scientist and CEO, to translate business requirements into models, analytical findings and project-ready solutions with communication and collaboration. **The analytics and dashboards built at Project 990 Inc. contributed to the organization winning the ARNOVA President's Award. Apart from work, I regularly write technical blogs on LLM architectures, LangChain, data engineering, and storage systems.**

EDUCATION

Indian Institute of Technology Delhi (IIT Delhi)	Feb 2026 – Present
Professional Certification Programme in FinTech	Online
Indiana University Bloomington	Aug 2023 – May 2025
Master of Science in Data Science	Bloomington, Indiana, USA
Indus University	Jul 2019 – Apr 2023
Bachelor of Technology in Computer Science	Ahmedabad, Gujarat, India

EXPERIENCE

Data Scientist	Jan 2025 – May 2026
Project 990 Inc.	Columbus, Ohio, USA
Implemented scalable ETL pipelines on Databricks to ingest financial grant data from 7 different sources spanning from 2013 to 2021, reducing processing time by 40% and automating structured data warehousing in Snowflake for multi year financial and operational analysis. Merged and validated datasets containing grant amounts and donor information using SQL. Transformed over 130,000 records using Python, applying predictive modeling and time series analysis to identify grant distribution trends that supported resource allocation and forecasting. Developed interactive Tableau dashboards to visualize donation trends, asset-to-grant distribution ratios, and funding disparities. Consulted with executive leadership to define KPIs and ensured data reproducibility by documenting pipelines and maintaining version control using Git.	
Machine Learning Intern	May 2024 – Aug 2024
ManekTech	Ahmedabad, Gujarat, India
Built a Python backend for AQI forecasting with automated data ingestion from REST APIs. Developed a React.js interface to monitor air quality trends, supporting technology implementation and real time operational tracking. Improved data preprocessing and feature selection for Linear Regression and Decision Tree models, and improved model accuracy by 7.6% through hyperparameter tuning and model evaluation techniques. - Enabled early alerts for hazardous conditions using data modeling techniques and deployed the trained model using Docker with custom monitoring scripts, integrating CI/CD pipelines for automated deployment and operational support.	
Data Analytics and Machine Learning Intern	Aug 2022 – Aug 2022
Start Tech Academy	Ahmedabad, Gujarat, India
Built a CNN-based image classification model and applied NLP techniques to analyze 2,000 image text pairs from educational materials. Optimized the solution using TensorFlow and Transformers, improving prediction accuracy by 11% for text classification tasks. Designed backend-to-frontend data pipelines to track model performance, support A/B testing, and deliver dashboard based insights for decision making. Developed Power BI dashboards with interactive visualizations for performance monitoring and reporting.	
Data Science Intern	Jun 2022 – Jul 2022
LetsGrowMore	Ahmedabad, Gujarat, India
Built a stacked LSTM model using TensorFlow and Keras to forecast stock closing prices from historical time series data. Used sliding window preprocessing, normalization, and multi-stock training to capture broader market patterns. Evaluated predictions using RMSE and MAE, and visualized predicted vs actual price trends with Matplotlib. Improved short-term forecasting accuracy by 12% through sequence length tuning and layer optimization to better capture price movement patterns.	

PROJECTS

Insightify - RAG Based AI Knowledge Assistant	GitHub
Designed an AI knowledge assistant with a full RAG pipeline to convert 10,000+ unstructured enterprise documents including PDFs, TXT and emails into searchable insights. Used LangChain for chunking, LangGraph for orchestration, and Sentence Transformers for embeddings. Created a FAISS vector index with metadata for fast semantic search. Integrated Hugging Face T5 models for context-aware answers, delivering 20% faster, more accurate queries on enterprise data.	

- Financial News Sentiment Tracker

GitHub

 - Built a financial news analytics pipeline using FastAPI, MongoDB, and FinBERT to process financial news articles and stock market data across AAPL, MSFT, TSLA, and NVDA. Reduced manual market tracking effort by 65% by aggregating stock movements into a unified daily analytics layer.
 - Implemented sentiment scoring and correlation analysis using custom statistical logic, improving sentiment classification by 32% across financial news headlines. Developed Streamlit dashboard to monitor price trends, sentiment movement, article volume, and ticker level market signals across multiple time windows.

- Fraud Pattern Detection in Digital Payments

GitHub

 - Built a real time fraud detection pipeline using Kafka, PySpark and XGBoost to process over 590,000 transactions. Created window based features for transaction velocity, user activity, and risk patterns. Improved short burst fraud detection by 22%.
 - Implemented fraud scoring with Isolation Forest, Statistical Rules, Graph Based risk signals and achieved 8% fraud detection coverage. Used NetworkX to identify fraud rings and burst patterns. Built a React Dashboard to display alerts, risk signals, and investigation timelines.

- Satellite Based Land Use Classification

GitHub

 - Built a satellite image classification system using ResNet18 with transfer learning to classify land patches into four categories including Agriculture, Forest, Urban, and Water. Mapped 10 raw dataset folders into final classes with a 224x224 input pipeline trained using Cross Entropy loss and Adam optimizer.
 - Achieved 96% test accuracy on unseen data using a CNN based evaluation pipeline and generated visual outputs including Grad-CAM visualizations, prediction and misclassification galleries, a confusion matrix, and a tile based overlay map that converts patch level predictions into a larger land use view.

SKILLS SUMMARY	
Programming Languages	: Python, R, SQL, C, C++, C#, HTML, CSS, LangChain, LangGraph
Machine Learning & AI	: LLMs, NLP, RAG, Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, Hugging Face Transformers, CNN
AI Tools & Platforms	: GitHub Copilot, Cursor, Vertex AI, Azure AI Search, Ollama, Microsoft Copilot, ChatGPT, Claude
Cloud & Data Platforms	: AWS (Lambda, S3, EC2, SageMaker), GCP, BigQuery, Snowflake, Databricks, Azure Data Lake
Data Engineering & Pipelines	: Apache Spark, Apache Beam, PySpark, Delta Lake, Airflow, Dagster, Azure Data Factory, dbt, Kafka, REST APIs
Databases & Storage	: PostgreSQL, MySQL, SQL Server, Spark SQL, NoSQL, ClickHouse, Pinecone, Chroma, MongoDB, Cassandra
Data Visualization & BI Tools	: Power BI, Tableau, Qlik, DAX, Looker Studio, Superset, Matplotlib, Seaborn, Adobe Analytics, Streamlit, Plotly
Analytics Tools	: Pandas, NumPy, Jupyter Notebook, Microsoft Excel, Google Sheets, VBA, Power Query
Infrastructure & DevOps	: Docker, Kubernetes, Azure DevOps, Git, CI/CD, MLOps, Unit Testing, Scrum, Linux

- CERTIFICATIONS
- AWS Data Engineer (AWS), Data Science Professional (Oracle), Data Science (IBM), SQL Fundamentals (DataCamp), Data Analytics (Cisco), Tableau Certification (Tableau), Microsoft Power BI Certification (Microsoft), Gen AI Certification (Indiana University), Prompt Engineering (DeepLearning.AI)

- AWARDS
- International Mathematics Olympiad (IMO) Gold Medalist
National Science Olympiad (NSO) Gold Medalist